



Outline of the lecture

The history of Informatics & HCI

Our roots and evolution

Conceptual Foundations

Information Processing | Gestalt Theory | Functional Approaches

Action Research & Praxeology

Work(place) studies & ethnography | Contextual Approaches |
Praxeology & Appropriation | Activity Theory

Structuration & Infrastructuring

A holistic look at Human-System-Interaction

HCI in Practice

Designing | Prototyping | Evaluation

Infrastructuring – How to Infrastructure

9 central characteristics of infrastructure: **relations**, not technologies; **emerging**, not designed



Embedded in other social and technological structures



Transparent in invisibly supporting work



Have a spatial and temporal reach or scope



Comprises taken-for-granted artifacts and organizational arrangements learned as part of membership



Normally invisible, become visible upon breakdown



Plug in other infrastructures and tools in a standardized way, and are modified by scope and conflicting (local) conventions



Shape and are shaped by the conventions of practice



Do not grow de novo but wrestle with the inertia of the installed based and inherit strengths and limits from that base



Fixed in modular increments, not all at once or globally

Infrastructuring – How to Infrastructure

9 central characteristics of infrastructure: **relations**, not technologies; **emerging**, not designed

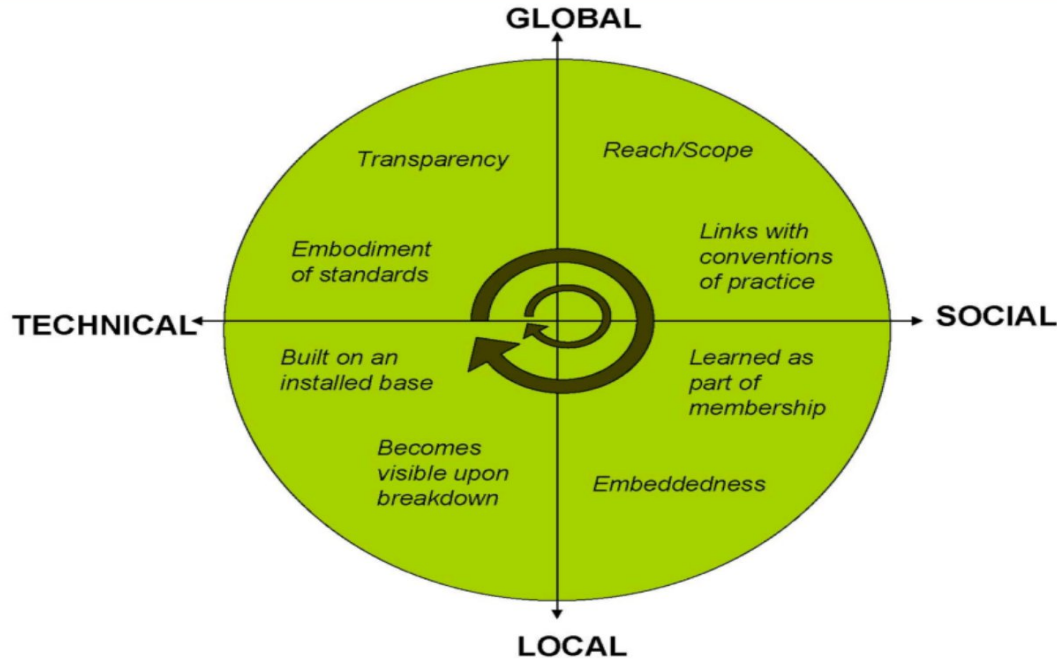
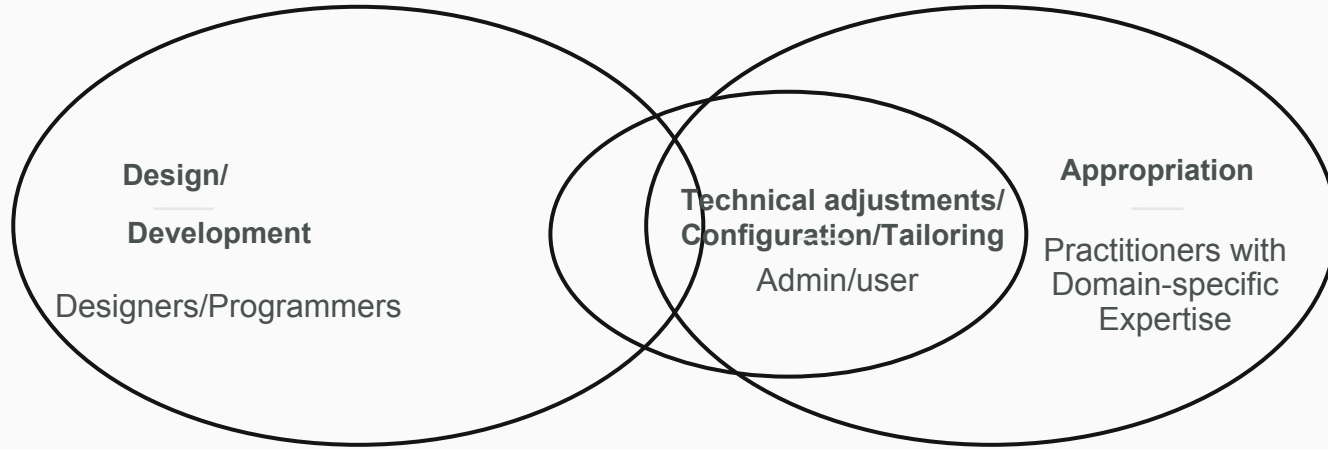


Figure 1. Cyberinfrastructure as distributions along technical/social & global/local axes (diagram courtesy of Florence Millerand).

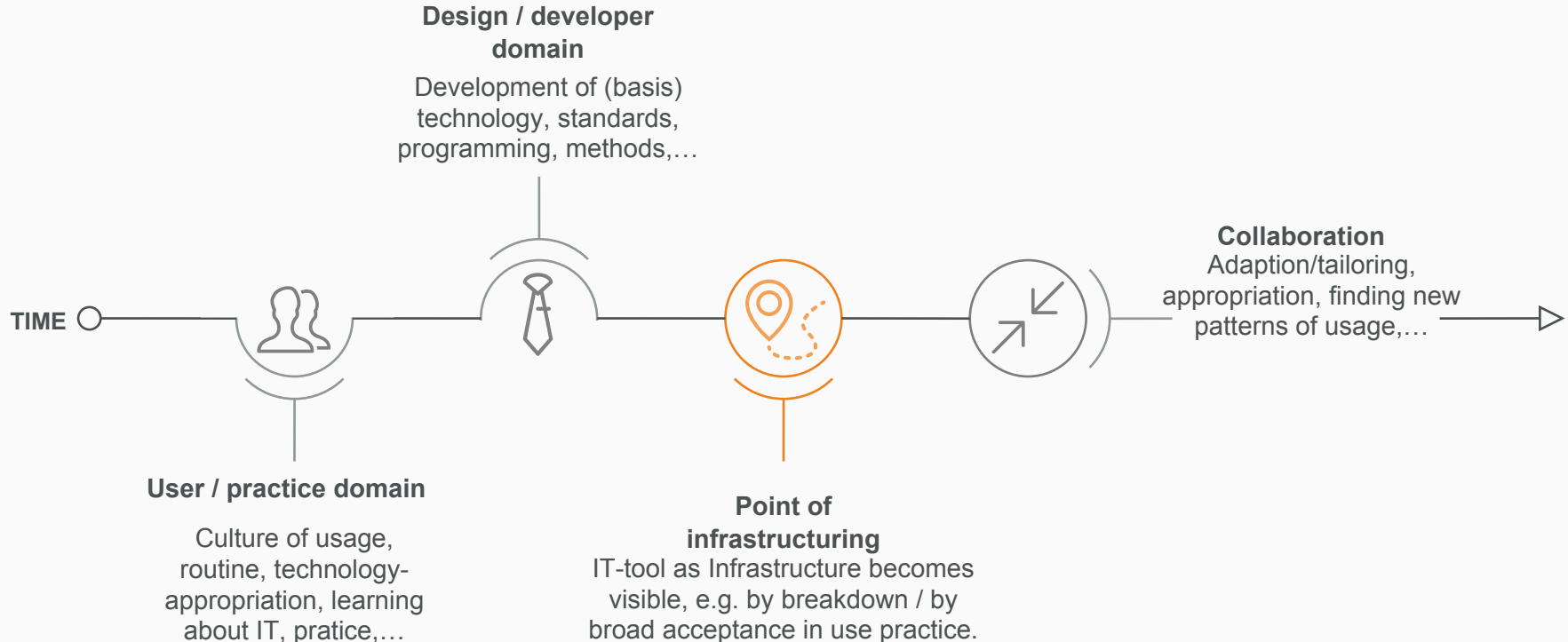
Creative activities which result in Usage

Not only designer/developer creativity (left), also practitioner creativity...



Infrastructuring – intended by practice

Infrastructuring as a holistic view of IT



Infrastructuring – one possible new theory

Case example: the discovery of a navigation-app as we know it today

Development of (multi-) touch (since the 70s!),
Microelectronics, the development of a navigation app



Infrastructuring: Overview of a Theory

